

Robert Schilling

SILICON ENGINEER · HARDWARE SECURITY

12 Titch Street, CB5 8JN Cambridge, UK

☎ (+44) 7356 019393 | 📞 (+43) 664 4496130 | ✉ schilling.ro@gmail.com | 🏠 www.schilling.casa | 🎮 Razer6 | 🕹 HackerOne: xanbanx

Summary

Silicon security architect specializing in platform security, silicon roots of trust, and hardware-software co-design.

Currently at Apple, working on security architecture for Apple silicon. Before joining Apple, I worked on platform security for custom silicon at Meta. Earlier, at Rivos Inc. (acquired by Meta), I led the development of Darjeeling, the SoC-integrated variant of OpenTitan, an open-source silicon root of trust — from design through tapeout, silicon bringup, and full provisioning.

Ph.D. from Graz University of Technology — hardware extensions and compiler support for fault-attack countermeasures on RISC-V processors. Active bug bounty researcher with [42 CVEs](#) discovered in GitLab. Passionate about automation and building reusable, scalable hardware-software co-designs that extend beyond security.

Work Experience

Apple

SoC SECURITY ARCHITECT

Cambridge, UK

Sept. 2026 - Present

- Define security architecture and requirements across Apple silicon platforms
- Collaborate with silicon and software teams to integrate and verify security throughout the SoC design flow

Meta

SILICON ENGINEER

Cambridge, UK

Feb. 2026 - Sept. 2026

- Platform security for Meta's custom silicon
- Advanced the RTL and design-verification maturity of [Caliptra](#), an open-source hardware root of trust, to support tapeout
- Defined a secure ROM patching architecture for post-silicon remediation of security vulnerabilities and functional defects

Rivos Inc. (acquired by Meta)

SECURITY ARCHITECT

Graz, Austria

Apr. 2023 - Feb. 2026

- Led development of [Darjeeling](#), the SoC-integrated OpenTitan Root of Trust
- FIPS 140-3 readiness for the security subsystem
- End-to-end silicon lifecycle: tapeout, bringup, and secure provisioning
- Upstreamed Darjeeling to the open-source OpenTitan project

Graz University of Technology

UNIVERSITY ASSISTANT

Graz, Austria

Apr. 2019 - Apr. 2023

- Research on secure code execution in the presence of physical attacks
- Teaching: Computer Organization and Networks, Digital System Design

Know-Center GmbH

RESEARCH ASSISTANT

Graz, Austria

Apr. 2016 - Mar. 2019

- Research on secure code execution in the presence of physical attacks

Graz University of Technology

STUDENT RESEARCHER

Graz, Austria

Feb. 2015 - Jul. 2015

- Built an AXI encryption pipeline for authenticated encryption, FPGA prototype with Linux boot

NXP Semiconductors Austria GmbH

SOFTWARE DEVELOPER

Gratkorn, Austria

May 2010 - Jan. 2015

- Lead development for automated measurement systems used for NFC-IC verification

Education

Graz University of Technology

PH.D. IN COMPUTER SCIENCE, PASSED WITH DISTINCTION

Graz, Austria

Apr. 2016 - Sept. 2023

- [Hardware Extensions and Compiler Support for Protection Against Fault Attacks](#)

ETH Zurich

MASTER THESIS

Zurich, Switzerland

Aug. 2015 - Feb. 2016

- Securing the Communication- and Memory-Interfaces of a Multi-Core Cluster

Skills

Security	Secure boot, attestation, DICE, confidential compute, key management & provisioning, FIPS 140-3, threat modeling, post-quantum cryptography, side-channel & fault mitigations
Hardware	RISC-V, OpenTitan, Caliptra, SoC integration, post-silicon validation, CDC/RDC, RAS, ASIC design & verification, FPGA prototyping
Programming	C, C++, Rust, Python, Tcl, Bash, SystemVerilog, VHDL, Assembly (RISC-V, ARM)
Tools	VCS, Xcelium, Bazel, LLVM, Git, CI/CD automation
Languages	German (native), English (fluent)

Scholarships & Grants

- 2015 **Scholarship of Excellence**, Industriellenvereinigung Kärnten
- 2015 **Scholarship of Excellence**, Graz University of Technology
- 2015 **Research Abroad Stipendium**, Graz University of Technology
- 2012 **Scholarship of Excellence**, Graz University of Technology
- 2011 **Scholarship of Excellence**, Graz University of Technology

Industry Talks

- [RWC'26] [Migrating a Silicon Root of Trust to Post-Quantum Crypto](#)
Amin Abdulrahman, Andrew Huang, **Robert Schilling**, et al.
Hardware-accelerated post-quantum crypto (ML-DSA, ML-KEM, SLH-DSA) on OpenTitan root of trust.
- [RISC-V'25] [OpenTitan Integrated: A RISC-V Open-Source Silicon Root-of-Trust for Large SoCs](#)
Robert Schilling, Samuel Ortiz, Ravi Sahita, and Andreas Kurth
Open-source silicon root of trust designed for integration into large SoC security subsystems.

Selected Publications

- [DIMVA'25] [FAULTLESS: Flexible and Transparent Fault Protection for Superscalar RISC-V Processors](#)
Moritz Waser, David Schrammel, **Robert Schilling**, and Stefan Mangard
3.5% LUT overhead on VeeR EH1; runtime-toggleable security/performance tradeoff; compatible with unmodified binaries.
- [COSADE'22] [FIPAC: Thwarting Fault- and Software-Induced Control-Flow Attacks with ARM Pointer Authentication](#)
Robert Schilling, Pascal Nasahl, and Stefan Mangard
Basic-block granular CFI with full LLVM compiler support; reuses existing ARM PAC hardware.
- [HASP'22] [SFP: Providing System Call Flow Protection against Software and Fault Attack](#)
Robert Schilling, Pascal Nasahl, Martin Unterguggenberger, and Stefan Mangard
OS-level protection ensuring syscall sequences can't be hijacked by fault or software attacks.
- [HOST'21] [SecWalk: Protecting Page Table Walks Against Fault Attacks](#)
Robert Schilling, Pascal Nasahl, Stefan Weiglhofer, and Stefan Mangard
Protects virtual-to-physical address translation against faults; covers multi-level page tables on application-class CPUs.
- [NDSS'20] [ConTeXt: A Generic Approach for Mitigating Spectre](#)
Michael Schwarz, Moritz Lipp, Claudio Canella, **Robert Schilling**, Florian Kargl, and Daniel Gruss
Minimal ISA extension marking secrets so they can enter but not transiently leak from registers; 0% overhead for non-secret code.
- [DATE'18] [High speed ASIC implementations of leakage-resilient cryptography](#)
Robert Schilling, Thomas Unterluggauer, Stefan Mangard, Frank K. Gürkaynak, Michael Muehlberghuber, and Luca Benini
Fresh re-keying as algorithmic DPA countermeasure; eliminates need for masking with minimal area overhead.
- [DATE'18] [Securing conditional branches in the presence of fault attacks](#)
Robert Schilling, Mario Werner, and Stefan Mangard
Protected comparisons linked to CFI scheme; prevents single-instruction skips on password checks, secure boot, privilege escalation.

- [ACSAC'18] [Pointing in the Right Direction - Securing Memory Accesses in a Faulty World](#)
Robert Schilling, Mario Werner, Pascal Nasahl, and Stefan Mangard
Compiler-generated pointer MACs detect fault-induced address corruption; protects loads and stores on embedded CPUs.

Additional Publications

- [DATE'23] [SCFI: State Machine Control-Flow Hardening Against Fault Attacks](#)
Pascal Nasahl, Martin Unterguggenberger, Rishub Nagpal, **Robert Schilling**, David Schrammel, and Stefan Mangard
Protects hardware FSMs against fault injection; prevents adversaries from skipping security-critical states.
- [CCS'23] [Multi-Tag: A Hardware-Software Co-Design for Memory Safety based on Multi-Granular Memory Tagging](#)
Martin Unterguggenberger, David Schrammel, Pascal Nasahl, **Robert Schilling**, Lukas Lamster, and Stefan Mangard
Overcomes small-tag collision limits of ARM MTE by combining object- and page-granular tags; gem5 + Linux + LLVM prototype.
- [HOST'21] [Protecting Indirect Branches Against Fault Attacks Using ARM Pointer Authentication](#)
Pascal Nasahl, **Robert Schilling**, and Stefan Mangard
Extends PAC-based protection to indirect jumps and function pointers in safety-critical embedded/automotive systems.
- [CCS'21] [CrypTag: Thwarting Physical and Logical Memory Vulnerabilities using Cryptographically Colored Memory](#)
Pascal Nasahl, **Robert Schilling**, Mario Werner, Jan Hoogerbrugge, Marcel Medwed, and Stefan Mangard
Encrypts memory with per-object keys; defeats both software exploits and physical probing without full memory safety overhead.
- [CCS'21] [HECTOR-V: A Heterogeneous CPU Architecture for a Secure RISC-V Execution Environment](#)
Pascal Nasahl, **Robert Schilling**, Mario Werner, and Stefan Mangard
TEE via dedicated fault-protected security core alongside untrusted application core; protects even against malicious OS.
- [VMCAI'19] [Small Faults Grow Up - Verification of Error Masking Robustness in Arithmetically Encoded Programs](#)
Anja F. Karl, **Robert Schilling**, Roderick Bloem, and Stefan Mangard
Shows that single-bit fault models underestimate real attacks; multi-bit faults can bypass arithmetic encoding.
- [DATE'19] [Protecting RISC-V Processors against Physical Attacks](#)
Mario Werner, **Robert Schilling**, Thomas Unterluggauer, and Stefan Mangard
First systematic study of fault countermeasures for RISC-V; instruction encoding, CFI, and memory protection.
- [TCAS'17] [An IoT Endpoint System-on-Chip for Secure and Energy-Efficient Near-Sensor Analytics](#)
Francesco Conti, **Robert Schilling**, Pasquale Davide Schiavone, Antonio Pullini, Davide Rossi, Frank Kagan Gürkaynak, Michael Muehlberghuber, Michael Gautschi, Igor Loi, Germain Haugou, Stefan Mangard, and Luca Benini
PULP-based SoC (FULMINE) with HW crypto; taped out in 65nm UMC; targets battery-powered IoT endpoints.
- [HiPEAC'17] [Multi-core data analytics SoC with a flexible 1.76 Gbit/s AES-XTS cryptographic accelerator in 65 nm CMOS](#)
Frank K. Gürkaynak, **Robert Schilling**, Michael Muehlberghuber, Francesco Conti, Stefan Mangard, and Luca Benini
Silicon-proven crypto accelerator for full-disk encryption integrated into PULP multi-core cluster.
- [CARDIS'17] [Leakage Bounds for Gaussian Side Channels](#)
Thomas Unterluggauer, Thomas Korak, Stefan Mangard, **Robert Schilling**, Luca Benini, Frank K. Gürkaynak, and Michael Muehlberghuber
Bridges gap between leakage-resilient crypto theory and real-world side-channel noise models.
- [FPL'17] [Transparent memory encryption and authentication](#)
Mario Werner, Thomas Unterluggauer, **Robert Schilling**, David Schaffenrath, and Stefan Mangard
Protects FPGA external memory against runtime probing and tampering; transparent to software.
- [Austrochip'14] [A low-area ASIC implementation of AEGIS128—A fast authenticated encryption algorithm](#)
Robert Schilling, Manuel Jelinek, Markus Ortoff, and Thomas Unterluggauer
First ASIC of AEGIS-128 (CAESAR competition candidate); area-optimized for constrained devices.